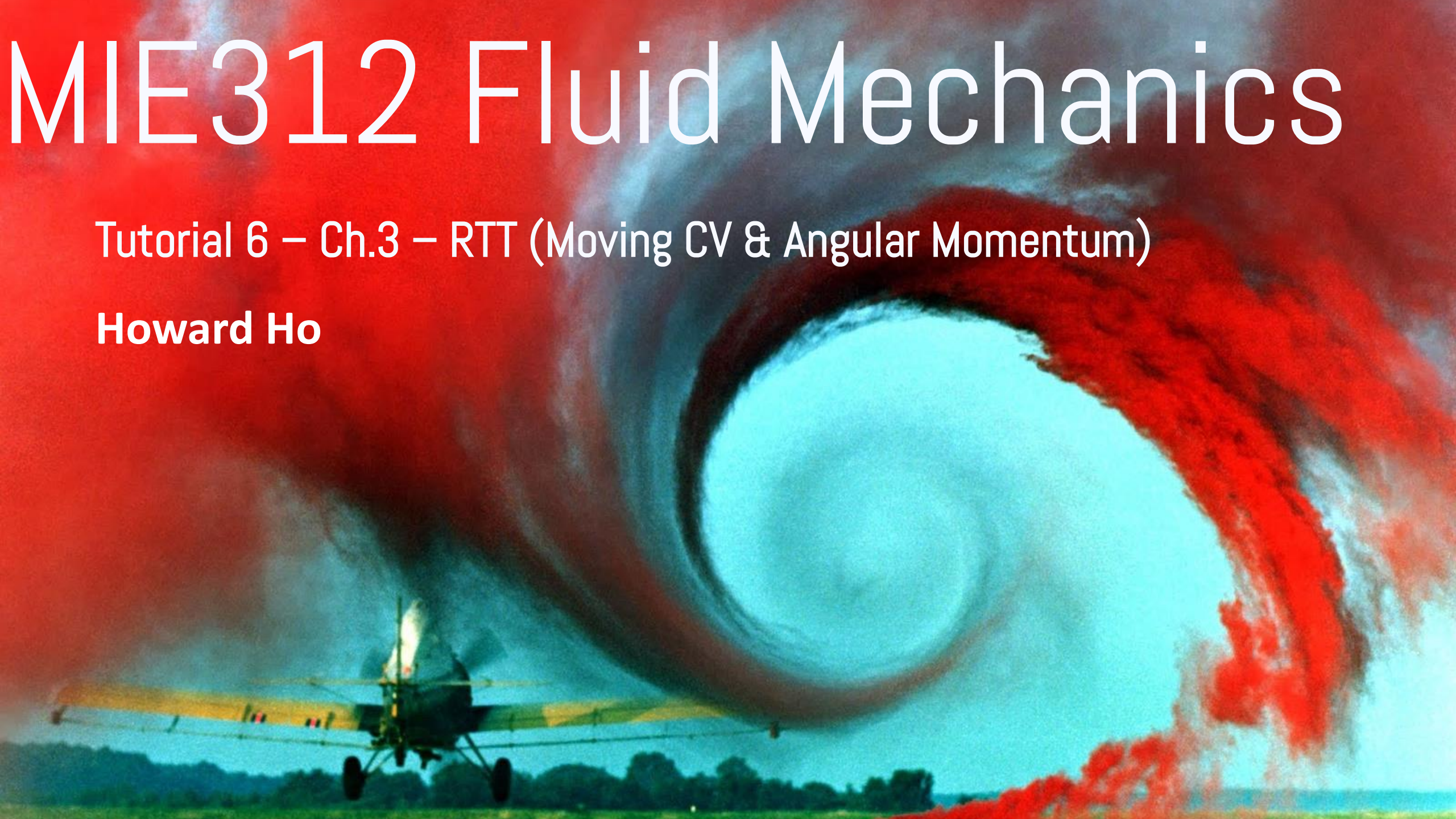
MIE312 Fluid Mechanics

Tutorial 6 – Ch.3 – RTT (Moving CV & Angular Momentum)

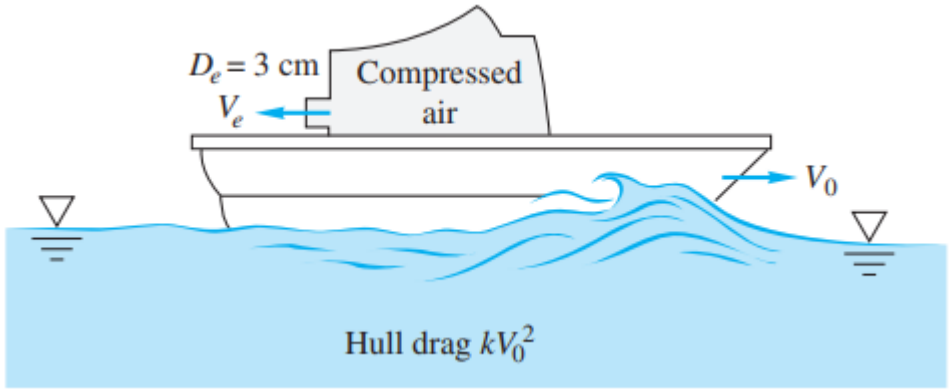
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Learning Objective

- Using RTT to solve moving CV problems
- Applying the angular momentum with RTT



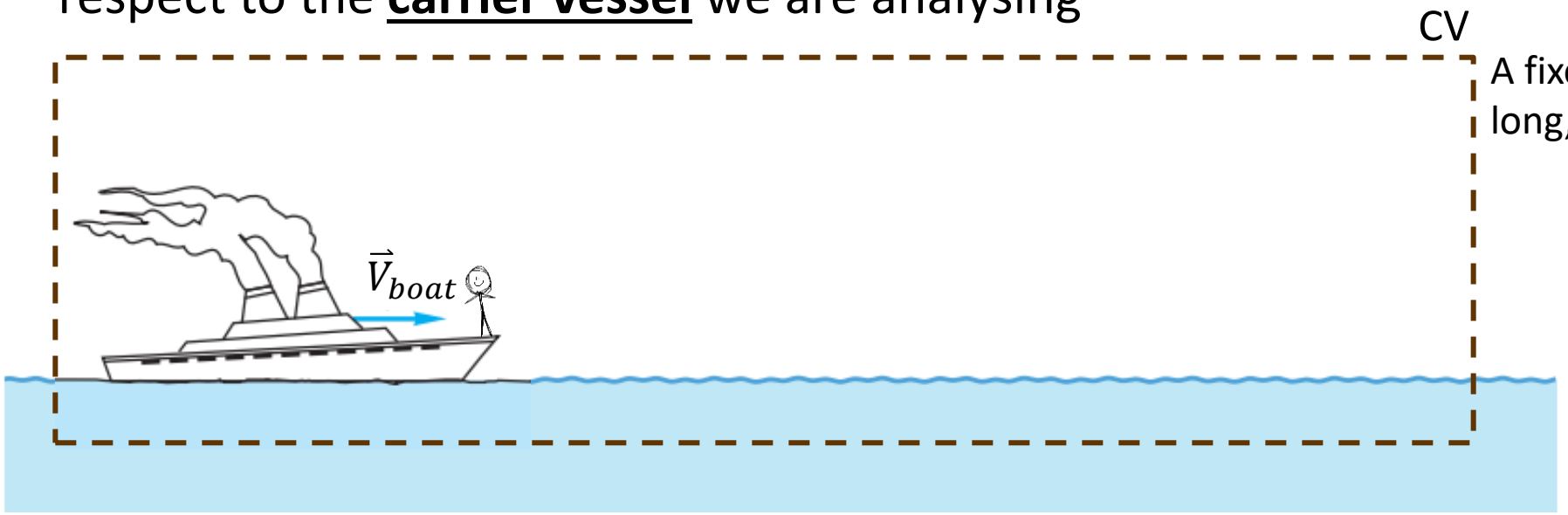




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Moving CV

- For moving CV, we want to apply relative motion so that the CV is no longer moving with respect to the **carrier vessel** we are analysing



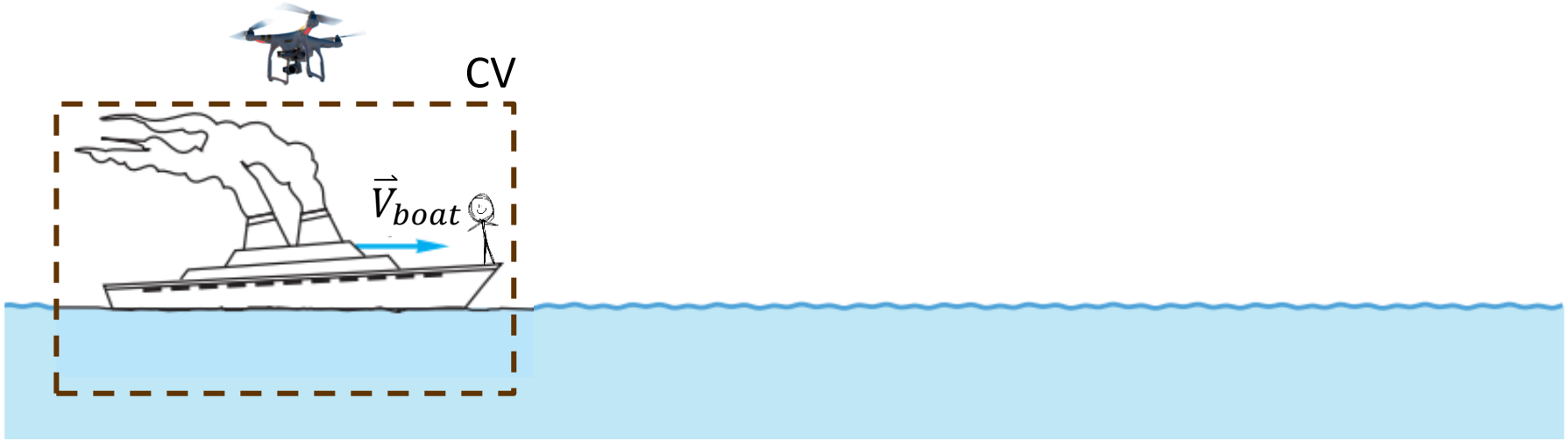
A fixed CV could be 5000km long, not very practical



Image we are flying a drone above the object of interest

The drone moves at the same speed as the boat

$$\vec{V}_{drone} = \vec{V}_{boat} = \vec{V}_{CV}$$



With a moving CV, we just need to cover roughly the boat's size

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Moving CV

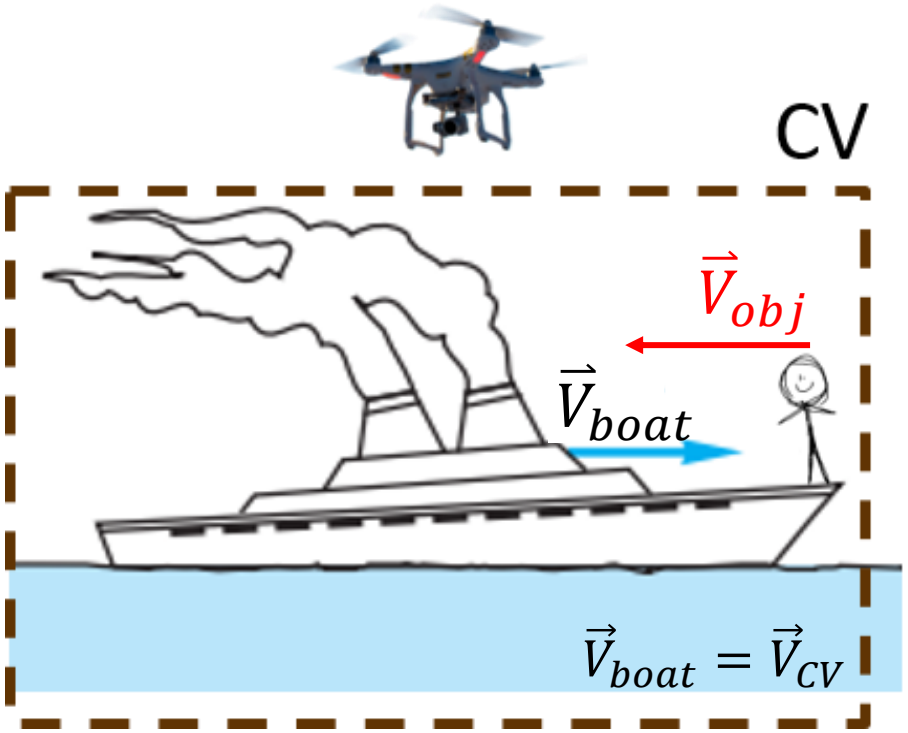
- Linear Momentum Transport equation:

$$\Sigma\vec{F} = \dot{m}\vec{V}_{out} - \dot{m}\vec{V}_{in}$$

$$\Sigma\vec{F} = \dot{m}_{obj/CV}\,\vec{V}_{obj/CV,\,out} - \dot{m}_{obj/CV}\,\vec{V}_{obj/CV,\,in}$$

We replace the velocity with relative velocity

$$\vec{V} = \vec{V}_{obj/CV}$$



Let the stick figure (Jack) be the object of interest:

$$\vec{V}_{obj/CV} = \vec{V}_{obj} - \vec{V}_{CV}$$

If:

$$\vec{V}_{obj} = \vec{V}_{CV}$$

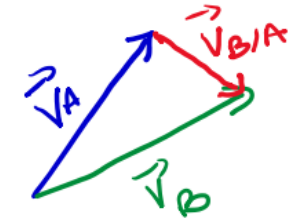
$$\vec{V}_{obj/CV} = 0$$

*But if $\vec{V}_{obj/CV} = 0$, the transport equation becomes zero
There'd be nothing to solve…

Recall:

$$\dot{m} = \rho A\vec{V}$$

Note: the $\dot{m}$ has to be positive by definition



Velocity

$$\vec{v}_{B/A} = \vec{v}_B - \vec{v}_A$$

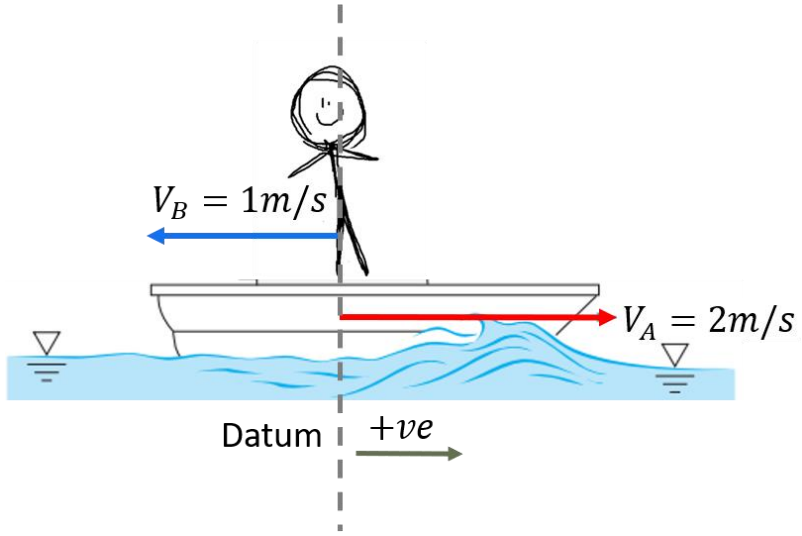
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Moving CV

- Linear Momentum Transport equation:

$$\Sigma\vec{F} = \dot{m}_{obj/CV}\,\vec{V}_{obj/CV,\,out} - \dot{m}_{obj/CV}\,\vec{V}_{obj/CV,\,in}$$

- Often, we need to determine which is $\vec{V}_{obj}$ and $\vec{V}_{CV}$

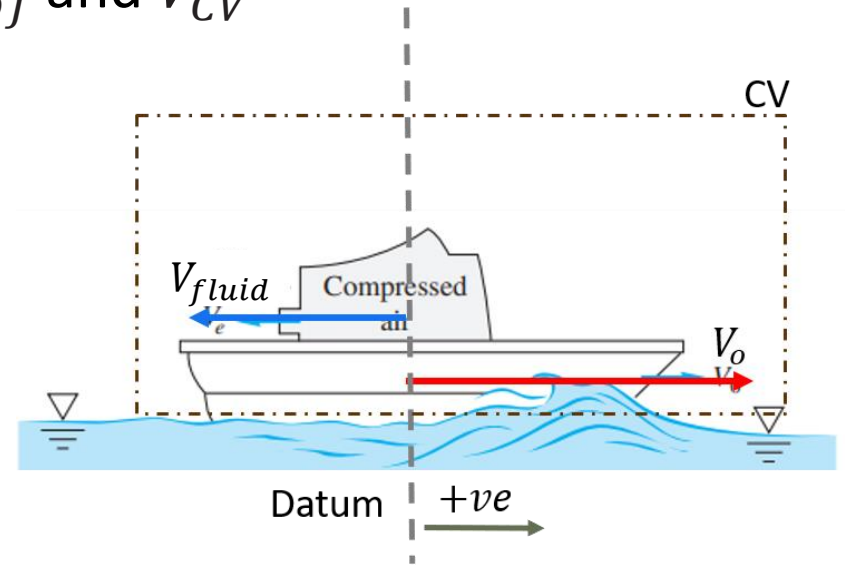


$\vec{V}_A$ is the carrier vessel ($\vec{V}_{CV} = \vec{V}_A$)

$\vec{V}_B$ is the object of interest ($\vec{V}_{obj} = \vec{V}_B$)

$$\vec{V}_{obj/CV} = \vec{V}_B - \vec{V}_A$$

$$\vec{V}_{obj/CV} = -1 - (2) = -3m/s$$



$$\vec{V}_{relative} = \vec{V}_{f/CV} = -V_f - (V_{CV})$$

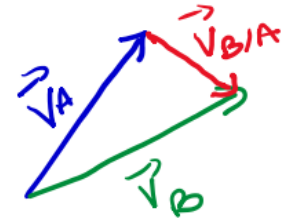
$$\Sigma\vec{F} = \dot{m}\vec{V}_{out} - \dot{m}\vec{V}_{in}$$

$$\Sigma\vec{F} = \dot{m}\vec{V}_{f/CV,\,out} - \dot{m}\vec{V}_{f/CV,\,in}$$

Note:

$\vec{V}_{obj} = \vec{V}_{fluid}$

$\vec{V}_o = \vec{V}_{CV}$



Velocity

$$\vec{v}_{B/A} = \vec{v}_B - \vec{v}_A$$

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RTT – Recap

- Reynolds Transport Theorem (RTT)

$$\frac{dB_{sys}}{dt} = \frac{d}{dt}\int_{CV} \rho\,\beta\,d\forall + \int_{Cs} \rho\,\beta\,\left(\overrightarrow{V_r}\cdot\hat{n}\right)dA$$

$$\beta = \frac{dB}{dm} \qquad \frac{dB_{sys}}{dt} = \frac{dB_{CV}}{dt} - \dot{B}_{in} + \dot{B}_{out}$$





- Conservation of mass

$$B = m, \quad \beta = \frac{dm}{dm} = 1$$

$$\frac{d}{dt}\rho\int d\forall = \dot{m}_{in} - \dot{m}_{out}$$



- Steady state,

$$\dot{m}_{in} = \dot{m}_{out}$$

- Linear Momentum Transport Equation

$$B = m\vec{V} \qquad \beta = \vec{V}$$

- Universal form

$$\Sigma\vec{F} = \frac{d}{dt}\int_{CV} \rho\vec{V}\,d\forall + \int_{CS} \rho\vec{V}\,\left(\overrightarrow{V_r}\cdot\hat{n}\right)\cdot dA$$

- Steady state and uniform inlet/outlet:

$$\Sigma\vec{F} = \dot{m}\vec{V}_{out} - \dot{m}\vec{V}_{in}$$

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Angular Momentum

- Reynolds Transport Theorem (RTT)

$$\frac{dB_{sys}}{dt} = \frac{d}{dt}\int_{CV} \rho\,\beta\,d\forall + \int_{Cs} \rho\,\beta\,(\vec{V_r}\cdot\hat{n})\,dA$$

$$\beta = \frac{dB}{dm} \qquad \frac{dB_{sys}}{dt} = \frac{dB_{CV}}{dt} - \dot{B}_{in} + \dot{B}_{out}$$



- Angular Momentum (H_o)

$$B = H_o = \int (r \times V)\,dm\,, \qquad \beta = \frac{dH_o}{dm} = r \times V$$

$r \times V$ is a cross product !

- Universal Form

$$\left.\frac{dH_o}{dt}\right|_{sys} = \frac{d}{dt}\left[\int_{CV} (r \times V)\rho\,d\forall\right] + \int_{CS} (r \times V)\rho(\vec{V_r}\cdot\hat{n})\,dA$$



- Non-deformable CV

$$\frac{dH_o}{dt} = \Sigma M_o = \Sigma(r \times F)_o$$

$$\Sigma \mathrm{M_o} = \frac{\partial}{\partial t}\left[\int_{CV} (r \times V)\,\rho d\forall\right] + \int_{CS} (r \times V)\,\rho(\vec{V_r}\cdot\hat{n})\,dA$$

- Steady state and uniform inlet/outlet:

$$\Sigma \mathrm{M_o} = \Sigma(r \times V)_{out}\dot{m}_{out} - \Sigma(r \times V)_{in}\dot{m}_{in}$$



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Angular Momentum Recap

- Consider a particle, P moving in the x-y plane
  - Position vector, $\vec{r}$
  - Velocity vector, $\vec{V}$
  - Angle between $\vec{r}$ and $\vec{V}$ is denoted as ϕ
- The angular momentum from the origin O:

$$\vec{H}_O = \vec{r} \times m\vec{V}$$

- If we use $\boldsymbol{r - \theta}$ coordinates for circular motion

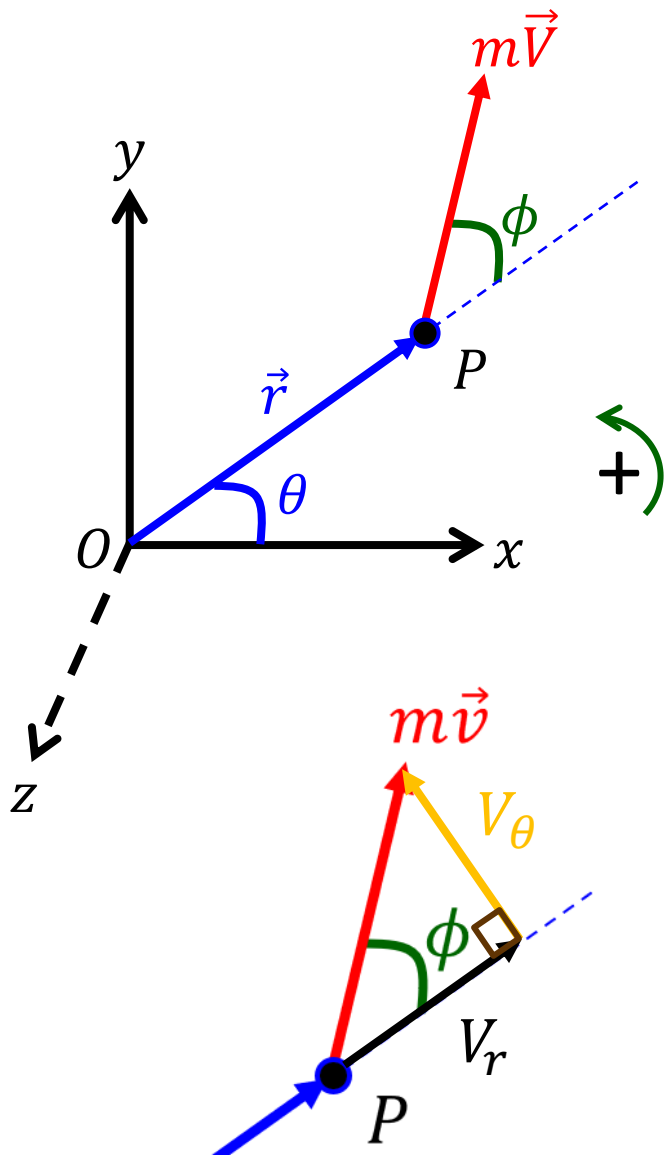
$$\vec{H}_O = \vec{r} \times m\vec{V} = mrV_\theta = mr^2\dot{\theta}$$

$$\vec{H}_O = mr^2\dot{\theta} = mr^2\omega$$

$\dot{\theta} = \omega$: angular velocity (rad/s)

- There's also a vector approach with $\hat{\imath}, \hat{\jmath}$ and $\hat{k}$ coordinates

*rpm is often provided instead of ω, need to convert those cases



SI units:

kg·(m/s) ·m = kg·m²/s = N·m·s

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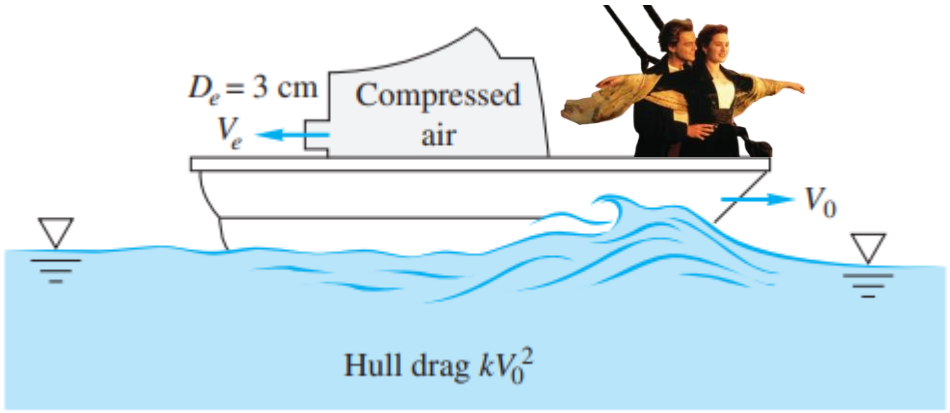
Question Time



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Problem 1

3.48 The small boat is driven at steady speed V_o by compressed air issuing from a 3-cm-diameter hole at $V_e = 343$ m/s and $p_e = 1$ atm, $T_e = 30°C$. Neglect air drag. The hull drag is kV_o^2, where $k = 19\ N \cdot s^2/m^2$. Estimate the boat speed V_o.



5 Steps Guide

1) Draw CV
2) Draw FBD
3) Mass Conservation
4) Momentum Equation
5) Break into x, y and z

3. CONSERVATION LAWS (CV)

RTT: $\frac{dB_{sys}}{dt} = \frac{\partial}{\partial t}\int_{CV}\rho\beta d\forall + \int_{CS}\rho\beta(\mathbf{V}_r\cdot\mathbf{n})dA$

Relative Vel: $\mathbf{V}_r = \mathbf{V}_f - \mathbf{V}_{cv}; \quad \dot{m}_{rel} = \rho A V_r$

Rotating outlet: $\mathbf{V} = \mathbf{V_r} - \mathbf{U}$; $\mathbf{U} = \boldsymbol{\omega}\times\mathbf{r}$

Steady 1-D Ang. Mom.: $M_z = \sum_{out}\dot{m}\,rV_\theta - \sum_{in}\dot{m}\,rV_\theta$

Mass: $\sum\dot{m}_{in} = \sum\dot{m}_{out} + \frac{\partial m_{cv}}{\partial t}$

Momentum: $\sum\mathbf{F} = \frac{\partial}{\partial t}\int\mathbf{V}\rho d\forall + \sum(\dot{m}\mathbf{V})_{out} - \sum(\dot{m}\mathbf{V})_{in}$

Ang. Mom (Stationary):

$\sum\mathbf{M}_o = \frac{\partial}{\partial t}\int(\mathbf{r}\times\mathbf{V})\rho d\forall + \sum(r\times\mathbf{V})_{out}\dot{m}_{out} - \sum(r\times\mathbf{V})_{in}\dot{m}_{in}$

Force of Jet on Vane:

$F_x = \dot{m}(V_{1,x} - V_{2,x}); \quad F_y = \dot{m}(V_{1,y} - V_{2,y})$

Energy: $\frac{P_1}{\gamma} + \alpha_1\frac{V_1^2}{2g} + z_1 + h_p = \frac{P_2}{\gamma} + \alpha_2\frac{V_2^2}{2g} + z_2 + h_t + h_L$

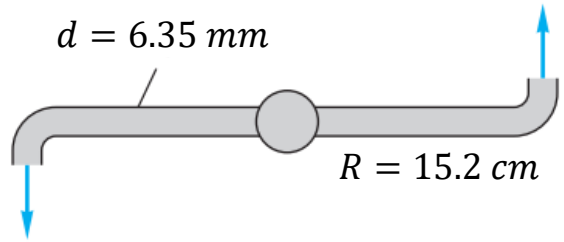
Factors: $\alpha = 2.0$ (Lam), $\alpha \approx 1.05$ (Turb)

Grade Lines: $EGL = HGL + \alpha V^2/2g$

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Problem 2



The horizontal lawn sprinkler has a water flow rate of 15.1 L/min introduced vertically through the centre. Estimate (a) the retarding torque required to keep the arms from rotating and (b) the rotation rate (r/min) if there is no retarding torque.

5 Steps Guide
1) Draw CV
2) Draw FBD
3) Mass Conservation
4) Momentum Equation
5) Break into x, y and z

3. CONSERVATION LAWS (CV)

RTT: $\frac{dB_{sys}}{dt} = \frac{\partial}{\partial t}\int_{CV}\rho\beta d\forall + \int_{CS}\rho\beta(\mathbf{V}_r\cdot\mathbf{n})dA$

Relative Vel: $\mathbf{V}_r = \mathbf{V}_f - \mathbf{V}_{cv}; \quad \dot{m}_{rel} = \rho A V_r$

Rotating outlet: $\mathbf{V} = \mathbf{V_r} - \mathbf{U}; \mathbf{U} = \boldsymbol{\omega}\times\mathbf{r}$

Steady 1-D Ang. Mom.: $M_z = \sum_{out}\dot{m}\,rV_\theta - \sum_{in}\dot{m}\,rV_\theta$

Mass: $\sum\dot{m}_{in} = \sum\dot{m}_{out} + \frac{\partial m_{cv}}{\partial t}$

Momentum: $\sum\mathbf{F} = \frac{\partial}{\partial t}\int\mathbf{V}\rho d\forall + \sum(\dot{m}\mathbf{V})_{out} - \sum(\dot{m}\mathbf{V})_{in}$

Ang. Mom (Stationary):

$\sum\mathbf{M}_o = \frac{\partial}{\partial t}\int(\mathbf{r}\times\mathbf{V})\rho d\forall + \sum(r\times\mathbf{V})_{out}\dot{m}_{out} - \sum(r\times\mathbf{V})_{in}\dot{m}_{in}$

Force of Jet on Vane:

$F_x = \dot{m}(V_{1,x} - V_{2,x}); \quad F_y = \dot{m}(V_{1,y} - V_{2,y})$

Energy: $\frac{P_1}{\gamma} + \alpha_1\frac{V_1^2}{2g} + z_1 + h_p = \frac{P_2}{\gamma} + \alpha_2\frac{V_2^2}{2g} + z_2 + h_t + h_L$

Factors: $\alpha = 2.0$ (Lam), $\alpha \approx 1.05$ (Turb)

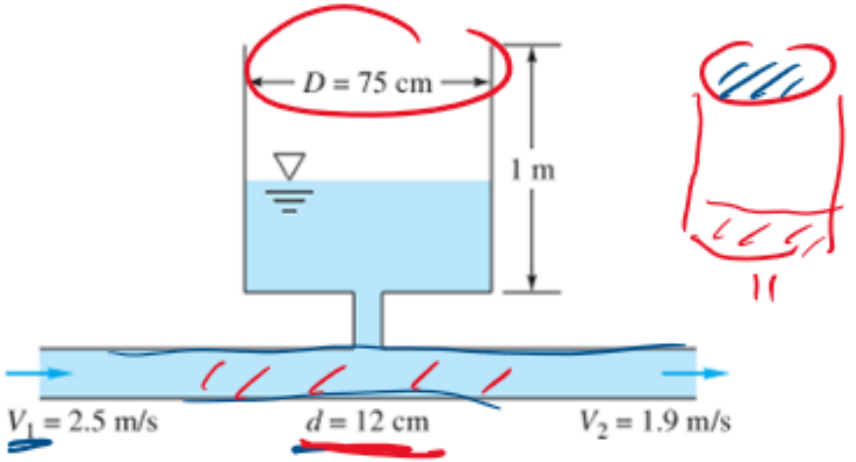
Grade Lines: $EGL = HGL + \alpha V^2/2g$

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Problem 3*

P3.12 The pipe flow in Fig. P3.12 fills a cylindrical surge tank as shown. At time $t = 0$, the water depth in the tank is 30 cm. Estimate the time required to fill the remainder of the tank.



$$\frac{d}{dt}\rho S\,dt = \dot{m}_{in} - \dot{m}_{out} \quad -(1)$$

$$\dot{m}_{in} = (\rho AV)_{in} = \rho A V_{in}$$

$$\dot{m}_{out} = \rho A V_{out}$$

(1) → (2)

$$\frac{d}{dt}\rho S\,dt = \rho A(V_{in} - V_{out})$$

$$\rho \frac{d}{dt} \forall = RHS$$

$$\rho \frac{d}{dt}(\pi R^2 h) = RHS$$

$$\rho \pi R^2 \frac{dh}{dt} = \rho A (V_{in} - V_{out})$$

R = 0.06 m, πr^2, h = 0.375 m

$$\frac{dh}{dt} = 0.01536 \ [m/s]$$

Find Δt,

$h_0 = 0.3$ m

$\therefore \Delta t = \frac{1 - 0.3}{0.01536}$

$\Delta t = 45.65$,

Solution: For a control volume enclosing the tank and the portion of the pipe below the tank,

$$\frac{d}{dt}\left[\int \rho\, dv\right] + \dot{m}_{out} - \dot{m}_{in} = 0$$

$$\rho\pi R^2 \frac{dh}{dt} + (\rho AV)_{out} - (\rho AV)_{in} = 0$$

$$\frac{dh}{dt} = \frac{4}{998(\pi)(0.75^2)}\left[998\left(\frac{\pi}{4}\right)(0.12^2)(2.5-1.9)\right] = 0.0153\ m/s,$$

$$\Delta t = 0.7/0.0153 = \mathbf{46}\,s \quad Ans.$$

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Practice Problems

Figure 3.15 shows a schematic of a centrifugal pump. The fluid enters axially and passes through the pump blades, which rotate at angular velocity ω; the velocity of the fluid is changed from V1 to V2 and its pressure from p1 to p2.

(a) Find an expression for the torque To that must be applied to these blades to maintain this flow.

(b) The power supplied to the pump would $P = \omega T_o$. To illustrate numerically, suppose $r_1 = 0.2m$, $r_2 = 0.5m$, and $b = 0.15m$. Let the pump rotate at 600rpm and deliver water at $2.5m^3/s$ with a density of $1000\ kg/m^2$. Compute the torque and power supplied.

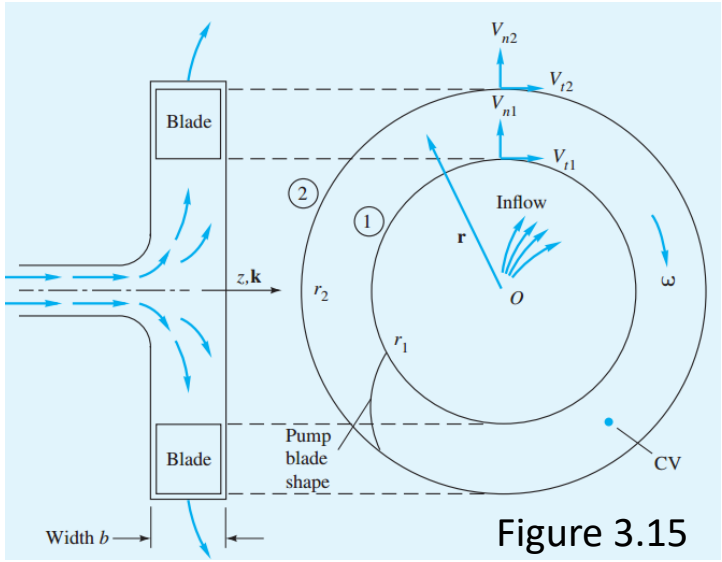
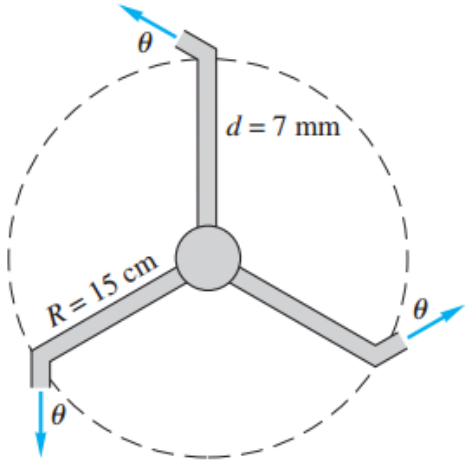


Figure 3.15

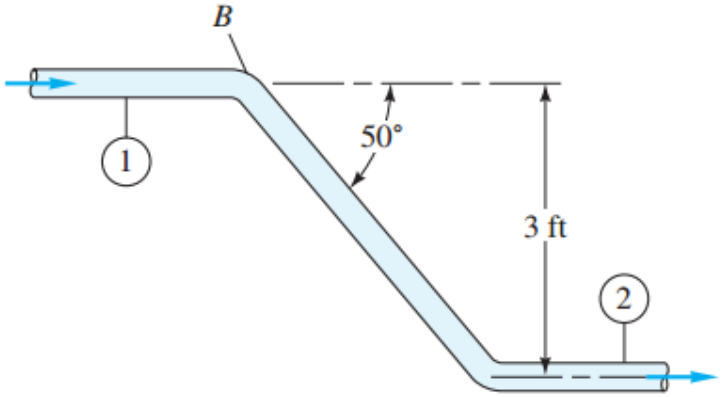
C3.2 A rigid tank of volume $\upsilon = 1.0\ m^3$ is initially filled with air at 20°C and po = 100 kPa. At time t = 0, a vacuum pump is turned on and evacuates air at a constant volume flow rate Q = 80 L/min (regardless of the pressure). Assume an ideal gas and an isothermal process. (a) Set up a differential equation for this flow. (b) Solve this equation for t as a function of (υ, Q, p, po). (c) Compute the time in minutes to pump the tank down to p = 20 kPa. [Hint: Your answer should lie between 15 and 25 minutes.]

P3.153 The three-arm lawn sprinkler of Fig. P3.153 receives 20°C water through the center at 2.7 m^3/h. If collar friction is negligible, what is the steady rotation rate in r/min for (*a*) $\theta = 0°$ and (*b*) $\theta = 40°$?

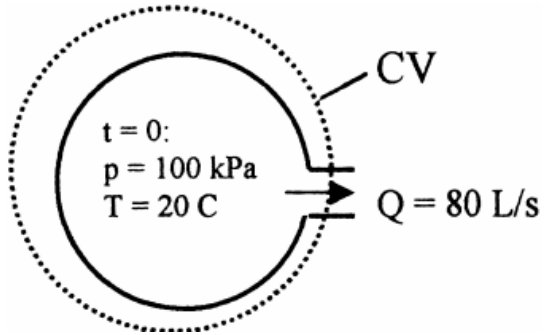


P3.153

P3.154 Water at 20°C flows at 30 gal/min through the 0.75-in-diameter double pipe bend of Fig. P3.154. The pressures are $p_1 = 30\ lbf/in^2$ and $p_2 = 24\ lbf/in^2$. Compute the torque *T* at point *B* necessary to keep the pipe from rotating.



P3.154



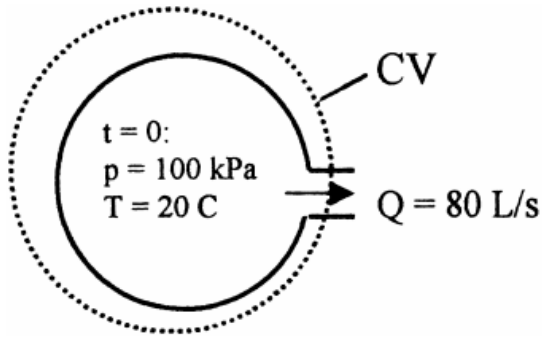
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Practice Problem Solution

$$\frac{d}{dt}\int \rho dV = \dot{m}_{in} - \dot{m}_{out}$$

$$\frac{d}{dt} m = \dot{m}_{in} - \dot{m}_{out}$$

C3.2 A rigid tank of volume $\upsilon = 1.0\ m^3$ is initially filled with air at 20°C and p_o = 100 kPa. At time t = 0, a vacuum pump is turned on and evacuates air at a constant volume flow rate Q = 80 L/min (regardless of the pressure). Assume an ideal gas and an isothermal process. (a) Set up a differential equation for this flow. (b) Solve this equation for t as a function of (υ, Q, p, p_o). (c) Compute the time in minutes to pump the tank down to p = 20 kPa. [Hint: Your answer should lie between 15 and 25 minutes.]



@ t=0s, P_o = 100 kPa

a) mass conservation

$$\frac{d}{dt}\int \rho\, d\forall = \dot{m}_{in} - \dot{m}_{out}$$

$$\frac{d\rho}{dt}\int d\forall = -\rho Q$$

$$\forall \frac{d\rho}{dt} = -\rho Q$$

$$\int \frac{1}{\rho}\frac{d\rho}{dt}dt = \int -\frac{Q}{\forall}\, dt$$

$$\int_{\rho_o}^{\rho} \frac{1}{\rho}\, d\rho = -\frac{Q}{\forall}\int_0^t dt$$

$$[\ln \rho]_{\rho_o}^{\rho} = -\frac{Q}{\forall} t$$

$$\ln \rho - \ln \rho_o = -\frac{Qt}{\forall}$$

$$\ln\left(\frac{\rho}{\rho_o}\right) = -\frac{Qt}{\forall}$$

Ideal Gas: $\frac{\rho}{\rho_o} = \frac{P}{P_o}$

$$\ln\left(\frac{P}{P_o}\right) = -\frac{Qt}{\forall}$$

$$t = -\frac{\forall}{Q}\ln\left(\frac{P}{P_o}\right)$$

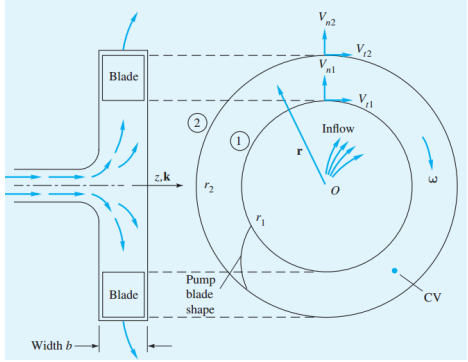
(c) $1m^3 = 1000\ L$

$\therefore 80\ L/min = 0.08 m^3/min$

$t \approx 20 min$,,

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Practice Problem Solution



Solution

The control volume is chosen to be the annular region between sections 1 and 2 where the flow passes through the pump blades (see Fig. 3.15). The flow is steady and assumed incompressible. The contribution of pressure to the torque about axis O is zero since the pressure forces at 1 and 2 act radially through O. Equation (3.59) becomes

$$\sum \mathbf{M}_o = \mathbf{T}_o = (\mathbf{r}_2 \times \mathbf{V}_2)\dot{m}_{\text{out}} - (\mathbf{r}_1 \times \mathbf{V}_1)\dot{m}_{\text{in}} \qquad (1)$$

where steady flow continuity tells us that

$$\dot{m}_{\text{in}} = \rho V_{n1} 2\pi r_1 b = \dot{m}_{\text{out}} = \rho V_{n2} 2\pi r_2 b = \rho Q$$

The cross product $\mathbf{r} \times \mathbf{V}$ is found to be clockwise about O at both sections:

$$\mathbf{r}_2 \times \mathbf{V}_2 = r_2 V_{t2} \sin 90° \, \mathbf{k} = r_2 V_{t2} \mathbf{k} \quad \text{clockwise}$$
$$\mathbf{r}_1 \times \mathbf{V}_1 = r_1 V_{t1} \mathbf{k} \quad \text{clockwise}$$

Equation (1) thus becomes the desired formula for torque:

$$T_o = \rho Q (r_2 V_{t2} - r_1 V_{t1}) \mathbf{k} \qquad \text{clockwise} \qquad \textit{Ans. (a) (2a)}$$

This relation is called *Euler's turbine formula*. In an idealized pump, the inlet and outlet tangential velocities would match the blade rotational speeds $V_{t1} = \omega r_1$ and $V_{t2} = \omega r_2$. Then the formula for torque supplied becomes

$$T_o = \rho Q \omega (r_2^2 - r_1^2) \quad \text{clockwise} \qquad (2b)$$

Convert ω to $600(2\pi/60) = 62.8$ rad/s. The normal velocities are not needed here but follow from the flow rate

$$V_{n1} = \frac{Q}{2\pi r_1 b} = \frac{2.5 \text{ m}^3/\text{s}}{2\pi(0.2 \text{ m})(0.15 \text{ m})} = 13.3 \text{ m/s}$$

$$V_{n2} = \frac{Q}{2\pi r_2 b} = \frac{2.5}{2\pi(0.5)(0.15)} = 5.3 \text{ m/s}$$

For the idealized inlet and outlet, tangential velocity equals tip speed:

$$V_{t1} = \omega r_1 = (62.8 \text{ rad/s})(0.2 \text{ m}) = 12.6 \text{ m/s}$$
$$V_{t2} = \omega r_2 = 62.8(0.5) = 31.4 \text{ m/s}$$

Equation (2*a*) predicts the required torque to be

$$T_o = (1000 \text{ kg/m}^3)(2.5 \text{ m}^3/\text{s})[(0.5 \text{ m})(31.4 \text{ m/s}) - (0.2 \text{ m})(12.6 \text{ m/s})]$$
$$= 33{,}000 \text{ (kg} \cdot \text{m}^2)/\text{s}^2 = 33{,}000 \text{ N} \cdot \text{m} \qquad \textit{Ans.}$$

The power required is

$$P = \omega T_o = (62.8 \text{ rad/s})(33{,}000 \text{ N} \cdot \text{m}) = 2{,}070{,}000 \text{ (N} \cdot \text{m)/s}$$
$$= 2.07 \text{ MW } (2780 \text{ hp}) \qquad \textit{Ans.}$$

In actual practice the tangential velocities are considerably less than the impeller-tip speeds, and the design power requirements for this pump may be only 1 MW or less.

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Practice Problem Solution

P3.153 The 3-arm lawn sprinkler of Fig. P3.153 receives 20°C water through the center at 2.7 m^3/hr. If collar friction is neglected, what is the steady rotation rate in rev/min for (a) $\theta = 0°$; (b) $\theta = 40°$?

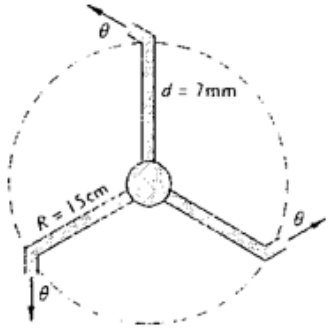


Fig. P3.153

Solution: The velocity exiting each arm is

$$V_o = \frac{Q/3}{(\pi/4)d^2} = \frac{2.7/[(3600)(3)]}{(\pi/4)(0.007)^2} = 6.50\ \frac{m}{s}$$

With negligible air drag and bearing friction, the steady rotation rate (Example 3.15) is

$$\omega_{final} = \frac{V_o \cos\theta}{R} \quad (a)\ \theta = 0°: \quad \omega = \frac{(6.50)\cos 0°}{0.15\ m} = 43.3\ \frac{rad}{s} = \mathbf{414}\ \frac{\mathbf{rev}}{\mathbf{min}} \quad \textit{Ans.}\ (a)$$

$$(b)\ \theta = 40°: \quad \omega = \omega_o \cos\theta = (414)\cos 40° = \mathbf{317}\ \frac{\mathbf{rev}}{\mathbf{min}} \quad \textit{Ans.}\ (b)$$

P3.154 Water at 20°C flows at 30 gal/min through the 0.75-in-diameter double pipe bend of Fig. P3.154. The pressures are $p_1 = 30$ lbf/in^2 and $p_2 = 24$ lbf/in^2. Compute the torque T at point B necessary to keep the pipe from rotating.

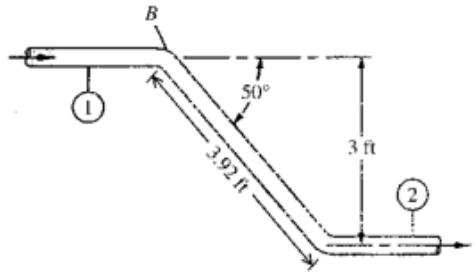


Fig. P3.154

Solution: This is similar to Example 3.13, of the text. The volume flow Q = 30 gal/min = 0.0668 ft^3/s, and $\rho = 1.94$ slug/ft^3. Thus the mass flow $\rho Q = 0.130$ slug/s. The velocity in the pipe is

$$V_1 = V_2 = Q/A = \frac{0.0668}{(\pi/4)(0.75/12)^2} = 21.8\ \frac{ft}{s}$$

If we take torques about point B, then the distance "h_1" from p. 143, = 0, and h_2 = 3 ft. The final torque at point B, from "Ans. (a)" on p. 143 of the text, is

$$T_B = h_2(p_2 A_2 + \dot{m} V_2) = (3\ ft)[(24\ psi)\frac{\pi}{4}(0.75\ in)^2 + (0.130)(21.8)] \approx \mathbf{40\ ft \cdot lbf} \quad \textit{Ans.}$$

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